

Course Code: CS3403		
Name of the Programme: B.Tech (H)		
Name of the Course: Network Security		
Semester: VI		
Course credit:	No. of hours per week (L+T+P):	Total no. of Teaching hours:
4	3:0:2	45L+30P = 75 hours
Course Objectives:		
a) Analyze network protocols (TCP, RIP, OSPF, RTP) and evaluate their security vulnerabilities along with protection mechanisms against common attacks.		
b) Understand and apply cybersecurity principles, including threats, vulnerabilities, defense strategies, and the role of cryptographic techniques like encryption, hashing, and digital signatures.		
c) Examine the interaction between QoS and network security to ensure prioritized, reliable, and secure data transmission across networks.		
d) Demonstrate practical skills in identifying, mitigating, and responding to network threats using simulation tools and hands-on lab exercises.		

Course outline (Syllabus of the course)

Syllabus:

Module - 1: Transmission Control Protocol (TCP)	No. of hours
	7
TCP Connection management, half-close, simultaneous open/close: ISN, Timeout, TCP Options, Path MTU: TCP State transitions: Reset segments, TCP Server operations: TCP Dataflow, interactive communication, Nagle Algorithm: TCP Flow control, window management. Various TCP congestion control protocols, DCCP, QUIC, DASH, and its relevance with respect to applications.	
Module - 2: Routing Protocols and SDN	No. of hours
	12
Distance-vector (RIP): Open Shortest Path First (OSPF): Router implementation and performance, Fabrics: Interdomain Routing- Global Internet, Routing areas: Interdomain Routing (BGP), AS relationships and policies: IPv6 Intro, Address space allocation, format, IP Multicast addresses and routing. Introduction to Software Defined Networking (SDN) and the recent trends on SDN.	

Module - 3: VPN and QoS	No. of hours
	8
Explicit routing and VPN tunnels: Mobile IP and routing among mobile devices: Remote Procedure Call: Real-Time Protocol (RTP) and RTCP: Congestion control and Queuing disciplines, RED: Quality of Service, Differentiated Services.	

Module - 4: Basics of Cryptography	No. of hours
	8
CIA Triad, Basics of cryptography -cryptographic hash functions – symmetric (DES, AES) and public-key encryption (RSA) -public key cryptography principles & algorithms - cipher block modes of operation - Secure Hash Functions (SHA512), Introduction to Post-Quantum Cryptography.	

Module - 5: Security Attacks	No. of hours
	10
Basics of Attacks, Network defense tools: Firewalls, VPNs, Intrusion Detection, and filters - Email privacy: Pretty Good Privacy and S/MIME - Network security protocols in practice-Introduction to Wireshark-SSL -IPsec, and IKE -DNS security- Secure Socket Layer and Transport Layer Security – Secure Electronic Transaction. Access control checks, Virus and Malware detection, Threat modelling, DDoS attacks, Network Monitoring and TCP Vs UDP, Routing Information Protocol (RIP), VLAN, VPN, NAT and VoIP	

Course Outcomes: After completing the course, the students will be able to:	
a.	Analyze the functionality of core transport and routing protocols, including TCP, RIP, and OSPF, to understand their roles in ensuring reliable and efficient network communication.
b.	Explain multimedia transport protocols and assess the significance of Quality of Service (QoS) in supporting diverse network applications.
c.	Understand and apply cryptographic algorithms to enable secure communication by ensuring data confidentiality, integrity, and authentication.
d.	Implement IP and Web security mechanisms to design secure network environments and address real-world cybersecurity challenges.

Text Books

1	L. Peterson and B. S. Davie, Computer Networks: A Systems Approach, 6th ed. Cambridge, MA, USA: Morgan Kaufmann, 2019. ISBN: 978-0123850591.
2	W. Stallings, Network Security Essentials: Applications and Standards, 6th ed. Noida, India: Pearson Education, 2021. ISBN: 978-9352866601.
3	W. Stallings, Cryptography and Network Security, 7th ed. Noida, India: PHI/Pearson, 2017. ISBN: 978-1-292-15858-7.

Reference books

1	K. R. Fall and W. R. Stevens, TCP/IP Illustrated, Volume 1: The Protocols, 5th ed. Boston, MA, USA: Pearson, 2012. ISBN: 978-0123850591.
2	L. Peterson, L. Brakmo, and B. S. Davie, TCP Congestion Control: A Systems Approach, ver. 1.1-Dev. [Online]. Available: https://tcpcc.systemsapproach.org/
3	J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach, 7th ed. Boston, MA, USA: Pearson, 2017. ISBN: 978-0-13-359414-0.

Lab Programs

[30 Hours]

PART A- SOFTWARE LAB

1	Introduction: Recap of Cisco Packet Tracer and network building using a simple network
2	Working principles and operation of NAT using CISCO packet tracer
3	Server and Client Interactions Using TCP Sockets
4	Encrypted data over TCP Socket using OpenSSL

PART B- HARDWARE LAB

(Using ICX7150 Ruckus switches and routers and WI-FI router)

	Introduction to Lab: Switch mode and Router mode of ICX7150 devices
5	IN SWITCH MODE- Introduction to hardware lab by using Telnet and simple ping between the same networks.
6	IN SWITCH MODE- Construct a VLAN network where communication between same VLAN should be possible and different VLANs should not be possible.

7.	IN ROUTER MODE- Construct an Inter-VLAN network and demonstrate communication between different VLANs.
8	IN ROUTER MODE- Demonstrate Access Control List in hardware lab.
9	IN ROUTER MODE- Configure Router on a Stick configuration using the Hardware lab.
10	IN ROUTER MODE- Demonstrate the working of OSPF.
11	DHCP snooping
12	Port-Security